

The Utility Problem

There are a variety of reasons why electric vehicles (EVs) have become more popular in recent years. One of the main reasons is the increasing concern about the environmental impact of traditional gasoline and diesel vehicles, as well as the desire to reduce dependence on fossil fuels. Additionally, advances in technology have led to the development of more efficient and affordable EVs, making them a more viable option for many people.

As the number of EVs increases and will keep increasing, it can have a number of impacts on the electric grid. Some of the key impacts include:

- Increased demand for electricity - As more EVs are charged, there will be an increased demand for electricity, particularly during peak hours and also in some locations with a higher probability of EV penetration (for example, family houses).
- Stress on the grid: If the increased demand for electricity is not managed properly, it can put stress on the grid, leading to power outages or brownouts.
- Need for more generation capacity - To meet the increased demand for electricity, electric utilities may need to invest in new generation capacity, such as solar or wind power, or expand existing power plants.
- Need for more transmission and distribution infrastructure - As more EVs are charged, there may be a need for more transmission and distribution infrastructure, such as power lines and substations, to move electricity from generation sources to where it is needed.
- Opportunities for grid balancing - On the other hand, with the growth of renewable energy sources and the implementation of smart charging systems and storage solutions, EVs can also provide opportunities for balancing the grid and making it more stable by absorbing surplus energy generated by renewable energy sources during off-peak hours.

It is a need for new infrastructure to support an increase in load. The infrastructure capacity levels, which are already strengthened by historically reduced investments in the grid, will definitely be tested by ongoing EV adoption.

Overall, the increased number of EVs on the road can have both challenges and opportunities for the electric grid, and the key is to have tools to tackle this challenge. The tools should allow for the estimation of increased load resulting from EV growth and also tools for identifying grid locations where the impact of increased load will be concerning.

The Data Solutions and Methods

While there are numerous methodologies for EV Load Growth Forecasting analysis, some of which are quite complex, the method which is described here aims to be simple and effective.

This methodology uses the current grid load, current EV penetration (households) and works with the prediction of EV adoption in the future. A combination of these data gives analysts information about increased feeder load caused by an increase in residential EVs. The case

scenario works with feeder-level data (consumers, EVs, load, etc.). In addition, this method puts increased feeder load into comparison with feeder or transformer capacity. The ability to segment the grid using feeder or transformers allow for a more granular analysis which can take into account variabilities in future EV adoption, such as high-density residential area or socioeconomic characteristics of neighbourhoods.

The following text will describe the process using the Awesense Energy Transition Platform. The process is described in detail so that the entire thought process is clear. With this, any analyst trying to reproduce this methodology will have the opportunity to adapt and modify it accordingly to meet her/his specific needs.

The number of residential meters:

- For the analysis, it is important to know the total number of residential meters (establishing the base):
 - This can be achieved by using `grid_get_downstream()` with further filtering parameters for meter element type with a residential attribute.
 - For the logical division of the analysis, this data can be fetched for each feeder in the examined area.
- Estimate the number of EV chargers in this grid. Estimation can be based on the following assumptions:
 - Current EV penetration in the US is 6%.
 - An analyst can use any other penetration rate which is common for his utility service area.
- A combination of these data will estimate the number of residential EVs for each feeder.

The future number of EVs and resulting increased load:

- To estimate the future penetration of EVs, the model might use the following assumptions¹:
 - The number of EVs will increase by between 5% to 34% per year.
 - The increase will be linear or compounded.
 - The total number of EVs will not exceed 1.88 vehicles per household.
 - Once the number of EVs reaches 1.88 vehicles per household, it will stay constant.
- Using the above assumptions, the number of EVs can be predicted up to the year 2050.
- To estimate the increase in load originating from EVs, the following assumptions can be made²:

¹ The mentioned assumptions are assumptions suited mostly for the US, analyst might want to use different figures.

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- The average American driver drives 20,000 km/year.
 - The average EV uses 0.2 kWh/km.
 - This results in an average of 4,000 kWh/year per EV.
 - Converting this energy requirement into average power means that each EV uses an average of 0.46 kW each hour of the day.
- The combination of the above-mentioned assumptions will yield an increase/contribution EV load for each year for each examined feeder.

To take the extra step, an analyst might want to put the increased load into the perspective of the current feeder load capacity. To do that, the following process might be put into practice.

Calculate the available load capacity per feeder or transformer:

- Aggregate the (kWh consumption) time series data to calculate the average power consumption per feeder or feeders.
 - This can be achieved by incorporating the function `ts_data_source_select()`.
- Determine the load capacity of the feeder or feeders:
 - While there might be multiple ways of determining the feeder load capacity, the described method uses a top-of-the-feeder's breaker *rating_a* amperage value to calculate the load capacity.
- Calculate the average available capacity per hour by subtracting the average hourly power consumption from the feeder's load capacity. This is the available capacity for the feeder. The available capacity for the (HV/MV) transformer will be the sum of all its feeders.

Further, the annual increase/contribution EV load can be applied to determine the feeders or transformers which will reach or exceed the capacity first. These feeders or transformers can be prioritized for investment to prevent harmful scenarios. Further analysis might be performed to show the effect of demand response where EV charging is shifted to a time of high available capacity (for example, during the night).

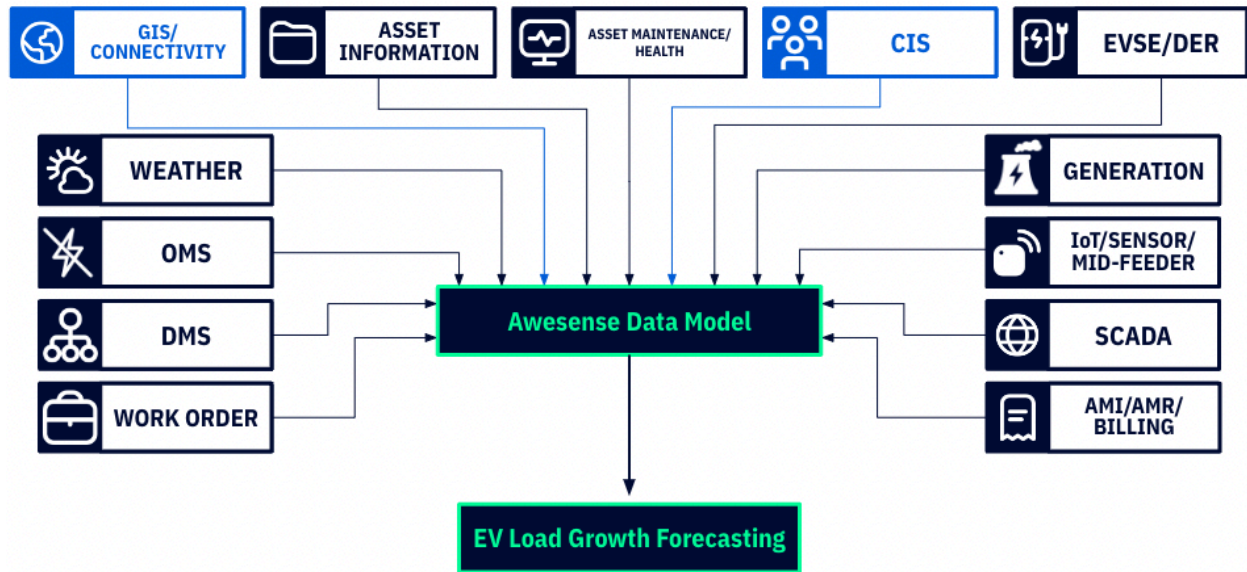
EV Load Growth Forecasting

[UC41-01 -] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub [link](#) to the Jupyter notebook.